## GLOBAL SUPERSTORE

# SALES ANALYSIS

*Professional Data Analytics Project Report*

Excel Data Preparation • Power Query • Power BI • SQL Validation

| Project component | Implementation |
|---|---|
| Data extraction | Excel |
| Data cleaning & feature engineering | Excel Power Query Editor |
| Data modeling & analytical measures | Power BI |
| Interactive dashboard | Power BI — 3 analytical pages |
| Independent validation | SQL |
| Analytical scope | Sales, profitability, customers, products, discount, shipping, geography and YoY |

*Prepared from the supplied project workbook, Power BI file, dashboard evidence and SQL validation script.*

# 1. Executive Summary

This project develops an end-to-end analytical workflow for the Global Superstore dataset. The prepared analytical dataset contains 51,255 records spanning 2011–2014 and supports 25,035 distinct orders, 4,873 customers, 147 countries, 13 regions and 2,875 product names.

The workflow is deliberately separated into preparation, analytical modeling, visualization and validation layers. Excel is used for extraction and working with the source data; Power Query provides repeatable cleaning and feature engineering; Power BI provides the dimensional model, measures and interactive multi-page dashboard; and SQL is used independently to reproduce selected business metrics and validate the dashboard outputs.

At the overall level, the cleaned data records $12.63M in sales, $1.46M in profit, an overall profit margin of 11.6%, 178,184 units, an average order value of $505, and average shipping time of 3.97 days.

Commercial performance increased consistently across the four years: sales rose from $2.26M in 2011 to $4.30M in 2014. The SQL validation logic independently calculates year-over-year sales and profit growth using annual aggregation and LAG(), providing a reproducible check against Power BI's time-based measures.

## Executive KPI Snapshot

| KPI | Value | Interpretation |
|---|---|---|
| Total Sales | $12.63M | Overall revenue generated in the cleaned analytical dataset |
| Total Profit | $1.46M | Net profit contribution across records |
| Profit Margin | 11.6% | Profit as a percentage of sales |
| Total Orders | 25,035 | Distinct order identifiers |
| Total Customers | 4,873 | Distinct customer identifiers |
| Average Order Value | $505 | Sales divided by distinct orders |
| Average Ship Days | 3.97 | Mean shipping duration at row level |
| Total Quantity | 178,184 | Units sold |

## 2. Report Structure

This report documents the project as a standalone analytical deliverable. The sections follow the actual implementation sequence and then move from dashboard interpretation to independent SQL validation.

| Section | Purpose |
|---|---|
| 1. Executive Summary | Project documentation |
| 2. Report Structure | Project documentation |
| 3. Project Definition and Business Questions | Project documentation |
| 4. Dataset and Source Structure | Project documentation |
| 5. End-to-End Analytical Workflow | Project documentation |
| 6. Excel Data Extraction and Working Structure | Project documentation |
| 7. Power Query Cleaning and Feature Engineering | Project documentation |
| 8. Data Quality Assessment | Project documentation |
| 9. Power BI Data Model | Project documentation |
| 10. Analytical Measures and DAX Framework | Project documentation |
| 11. Power BI Dashboard Architecture | Project documentation |
| 12. Dashboard Page 1 | Overview |
| 13. Dashboard Page 2 | Sales Analysis |
| 14. Dashboard Page 3 | Profit Analysis |
| 15. Business Performance Analysis | Project documentation |
| 16. Product and Category Analysis | Project documentation |
| 17. Customer and Segment Analysis | Project documentation |
| 18. Shipping and Operational Analysis | Project documentation |
| 19. Geographic Analysis | Project documentation |
| 20. Discount and Profitability Analysis | Project documentation |
| 21. Year-over-Year Performance | Project documentation |
| 22. SQL Independent Validation Framework | Project documentation |
| 23. Validation Results and Reconciliation | Project documentation |
| 24. Key Business Insights | Project documentation |
| 25. Recommendations | Project documentation |
| 26. Limitations and Analytical Controls | Project documentation |
| 27. Conclusion | Project documentation |
| 28. Technical Skills Demonstrated | Project documentation |
| Appendix A | Data Dictionary |
| Appendix B | SQL Query Inventory |
| Appendix C | Evidence Register |

# 3. Project Definition and Business Questions

The project is designed to convert a transaction-level Global Superstore dataset into a decision-oriented analytical product. The focus is not only on producing charts, but on establishing a traceable pipeline from source data to validated business metrics.

## Business Questions

- How large is the business in terms of sales, profit, orders, customers and quantity?
- How have sales and profit changed over time, and what does year-over-year growth indicate?
- Which categories, sub-categories and products generate the strongest sales and profit?
- Which segments contribute most to revenue and profitability?
- How does discount intensity affect sales, profit and margin?
- How is shipping distributed across shipping modes and shipment-status classes?
- Which regions and countries contribute the greatest sales and profit?
- Where are the most important loss-making product opportunities?
- Can key Power BI metrics be independently reproduced in SQL?

## Project Objectives

1. Prepare a reliable analytical dataset from the Excel source.
2. Perform cleaning and feature engineering in Power Query so transformations remain traceable and refreshable.
3. Build an analytical model in Power BI using fact and dimension structures.
4. Create reusable measures for core KPIs, prior-year comparisons and growth analysis.
5. Design a multi-page dashboard that separates overview, sales and profit analysis.
6. Use SQL as an independent validation layer rather than relying solely on dashboard calculations.
7. Translate quantitative findings into practical business recommendations.

# 4. Dataset and Source Structure

The supplied workbook contains four worksheets: Master_Data, Cleaned_Data and Dashboard. The analytical work is centered on the cleaned dataset, while Master_Data provides the source-stage working structure.

| Workbook object | Observed structure | Role in project |
|---|---|---|
| Master_Data | 51,290 rows × 27 columns | Source-stage dataset used before final cleaning |
| Cleaned_Data | 51,255 rows × 32 columns | Prepared analytical dataset used for downstream analysis |
| Dashboard | Worksheet present | Workbook dashboard area; the principal interactive dashboard is delivered in Power BI |

Database Source: Kaggle — Global Superstore Dataset by Fatih Ilhan.

## Dataset Coverage

| Dimension | Count / range |
|---|---|
| Records | 51,255 |
| Order IDs | 25,035 |
| Customers | 4,873 |
| Products | 2,875 |
| Product IDs | 10,292 |
| Categories | 3 |
| Sub-categories | 17 |
| Countries | 147 |
| Regions | 13 |
| Segments | 3 |
| Shipping modes | 4 |
| Years | 2011–2014 |

## Core Analytical Fields

| Field group | Analytical purpose |
|---|---|
| Order_Date / Ship_Date | Transaction timing and shipping duration |
| Sales / Profit | Primary commercial and profitability measures |
| Quantity | Unit volume |
| Discount | Discount intensity |
| Category / Sub_Category / Product_Name | Product hierarchy |
| Customer_ID / Customer_Name / Segment | Customer analysis |
| Country / Region / Market | Geographic analysis |
| Ship_Mode / Ship_Days / Ship_Status | Operational analysis |
| Year / Quarter / Month_No / Month Name | Time analysis |
| Discount_Range / Discount_Category | Derived discount grouping |
| Profit_Margin_Percent / Profit_Status | Derived profitability indicators |

# 5. End-to-End Analytical Workflow

The project uses a layered workflow in which each technology has a defined responsibility. This separation reduces calculation duplication and creates a clear audit trail from source preparation to final validation.

| Stage | Tool | Primary activities | Output |
|---|---|---|---|
| 1 | Excel | Source extraction and initial workbook organization | Master_Data |
| 2 | Power Query | Cleaning, type standardization, duplicate removal, text cleanup and feature engineering | Cleaned_Data |
| 3 | Power BI | Dimensional modeling and relationship-driven analysis | Analytical semantic model |
| 4 | Power BI | Reusable measures, prior-year logic and dashboard visuals | Interactive 3-page dashboard |
| 5 | SQL | Independent KPI, trend, product, customer, shipping, geography and YoY validation | Validation results |
| 6 | Analysis | Interpretation and recommendations | Business conclusions |

The workflow is intentionally complementary: Power Query prepares the data, Power BI performs interactive analysis, and SQL provides an independent computational check.

## Traceability Principle

- Source-level changes are handled in the preparation layer.
- Business calculations used repeatedly in the dashboard are represented as Power BI measures.
- Dashboard pages focus on decision questions rather than exposing every technical calculation.
- SQL validation reproduces selected outputs independently from the Power BI visual layer.

# 6. Excel Data Extraction and Working Structure

The supplied workbook contains a source-stage Master_Data sheet with 51,290 rows and 27 columns and a prepared Cleaned_Data sheet with 51,255 rows and 32 columns. This establishes a clear distinction between source-stage data and the analytical table used downstream.

## Source-to-Analytical Progression

| Metric | Master_Data | Cleaned_Data | Change |
|---|---|---|---|
| Rows | 51,290 | 51,255 | -35 |
| Columns | 27 | 32 | +5 |
| Composite duplicates | 35 | 0 | Removed |

The 35-row reduction corresponds to duplicate Order–Product–Customer combinations observed in the source-stage data. The cleaned analytical table contains no remaining duplicates for that composite key.

## Extraction Controls

- The analytical table preserves transaction-level granularity.
- Date fields are retained for time intelligence.
- Identifiers remain available for distinct-order and distinct-customer calculations.
- Commercial, operational, customer, product and geographic attributes are retained for multidimensional analysis.

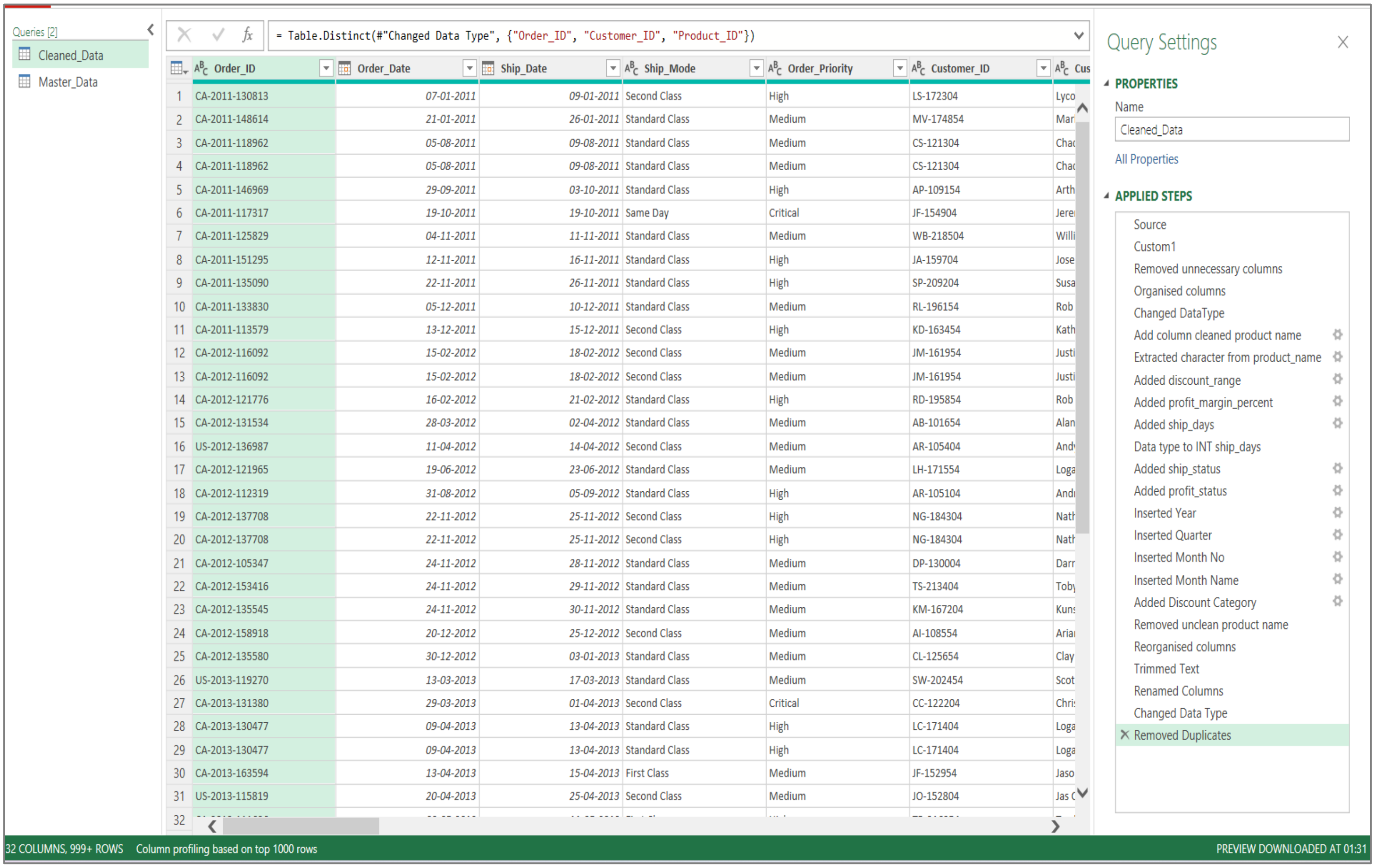
# 7. Power Query Cleaning and Feature Engineering

The supplied Power Query evidence shows a dedicated Cleaned_Data query with a sequence of applied steps. The transformation chain includes source acquisition, column reduction and organization, data-type changes, text cleanup, derived business fields and duplicate removal.

| Transformation sequence | Observed Power Query step |
|---|---|
| 1 | Source |
| 2 | Custom1 |
| 3 | Removed unnecessary columns |
| 4 | Organized columns |
| 5 | Changed DataType |
| 6 | Added column cleaned product name |
| 7 | Extracted character from product_name |
| 8 | Added discount_range |
| 9 | Added profit_margin_percent |
| 10 | Added ship_days |
| 11 | Data type to INT ship_days |
| 12 | Added ship_status |
| 13 | Added profit_status |
| 14 | Inserted Year |
| 15 | Inserted Quarter |
| 16 | Inserted Month No |
| 17 | Inserted Month Name |
| 18 | Added Discount Category |
| 19 | Removed unclean product name |
| 20 | Reorganized columns |
| 21 | Trimmed Text |
| 22 | Renamed Columns |
| 23 | Changed Data Type |
| 24 | Removed Duplicates |

## Feature Engineering Layer

| Derived field | Analytical role |
|---|---|
| Discount_Range | Groups discount values into interpretable ranges for impact analysis |
| Discount_Category | Provides a higher-level discount classification |
| Profit_Margin_Percent | Measures profitability relative to sales |
| Ship_Days | Converts order/ship dates into shipping duration |
| Ship_Status | Classifies shipping performance |
| Profit_Status | Classifies records by profitability outcome |
| Year | Supports annual trend and YoY analysis |
| Quarter | Supports quarterly performance analysis |
| Month_No / Month Name | Supports ordered monthly trend analysis |
| Cleaned Product Name | Improves product-name consistency for analysis |




*Power Query evidence supplied with the project — Cleaned_Data applied steps and 32-column analytical structure.*

# 8. Data Quality Assessment

Data quality was reviewed using both the prepared workbook and the SQL validation script. The checks focus on duplicates, date consistency, shipping duration, zero-sales records and completeness of key measures.

| Quality check | Result | Assessment |
|---|---|---|
| Source composite duplicates | 35 | Duplicate Order–Product–Customer combinations identified at source stage |
| Clean composite duplicates | 0 | No remaining composite duplicates |
| Order date after ship date | 0 | No invalid date ordering observed |
| Negative shipping days | 0 | No negative shipping duration observed |
| Zero-sales records | 1 | One exceptional transaction requires contextual treatment |
| Missing Profit_Margin_Percent | 1 | One missing value; associated with zero sales |
| Sales missing | 0 | No missing sales values |
| Profit missing | 0 | No missing profit values |
| Quantity missing | 0 | No missing quantity values |
| Ship_Days missing | 0 | No missing shipping-duration values |

The single missing profit-margin value is consistent with the presence of one zero-sales record. A conventional Profit ÷ Sales calculation is undefined for that row, so treating the margin as missing is analytically safer than forcing an artificial percentage.

## Quality Control Interpretation

- Duplicate control is effective at the selected transaction key.
- Date and shipping-duration checks support the integrity of the derived Ship_Days field.
- The zero-sales exception should remain visible in data-quality documentation rather than being silently converted into a percentage.
- Derived fields should be refreshed from Power Query rather than manually overwritten in the final analytical table.

# 9. Power BI Data Model

The supplied PBIX model structure contains a fact table, multiple dimensions and dedicated supporting tables. The model diagram identifies Fact_Sales alongside Dim_Date, Dim_Product, Dim_Customer, Dim_Geography and Dim_Ship, with a dedicated _Measures table and supporting Temp_Table and Last Refresh objects.

| Model object | Observed role |
|---|---|
| Fact_Sales | Central transaction-level analytical table |
| Dim_Date | Time-analysis dimension |
| Dim_Product | Product and category analysis |
| Dim_Customer | Customer and segment analysis |
| Dim_Geography | Country and regional analysis |
| Dim_Ship | Shipping-mode and operational analysis |
| _Measures | Dedicated measure layer for reusable calculations |
| Temp_Table | Supporting table used by report elements |
| Last Refresh | Refresh-status support object shown in the model/report |

The model organization separates descriptive attributes from the transactional fact table and provides a dedicated measure layer. This is appropriate for an interactive BI report because the same measures can be reused across cards, charts, tables and filtered views.

## Modeling Principles Applied

- Separate date analysis from transaction-level fields through Dim_Date.
- Centralize reusable measures in _Measures.
- Provide dedicated dimensions for product, customer, geography and shipping analysis.
- Keep transaction-level facts in Fact_Sales.
- Use report-level filters and dimension attributes to drive interactive slicing across pages.

# 10. Analytical Measures and DAX Framework

The PBIX contains a dedicated _Measures layer with reusable metrics covering core KPIs, prior-year values and year-over-year growth. The observed measure inventory includes the following.

| Measure group | Observed measures |
|---|---|
| Core KPIs | Total Sales, Total Profit, Profit Margin %, Total Orders, Total Customers, Total Ship Cost, Average Discount, Average Order Value, Average Ship Days |
| Prior-year measures | PY Sales, PY Profit, PY Customers, PY Orders, PY AVG Order Value, PY AVG Ship Days, PY Profit Margin |
| YoY measures | YoY Sales Growth %, YoY Profit Growth %, YoY Customers, YoY Orders, YoY AVG Order Value, YoY AVG Ship Days, YoY Profit Margin |

## Measure Design Principles

- Sales and profit are aggregated from the transaction fact.
- Orders and customers are treated as distinct identifiers rather than raw row counts.
- Average Order Value is based on sales divided by distinct orders.
- Profit Margin is expressed as profit relative to sales.
- Prior-year measures provide a comparison base for KPI cards and trend analysis.
- YoY measures express directional change rather than replacing the underlying absolute values.

The PBIX package also contains an extracted DAX query showing an evaluation of a sales aggregation over Fact_Sales. The report therefore documents the measure architecture from the model metadata and report layout without inventing formulas that are not directly exposed by the supplied PBIX package.

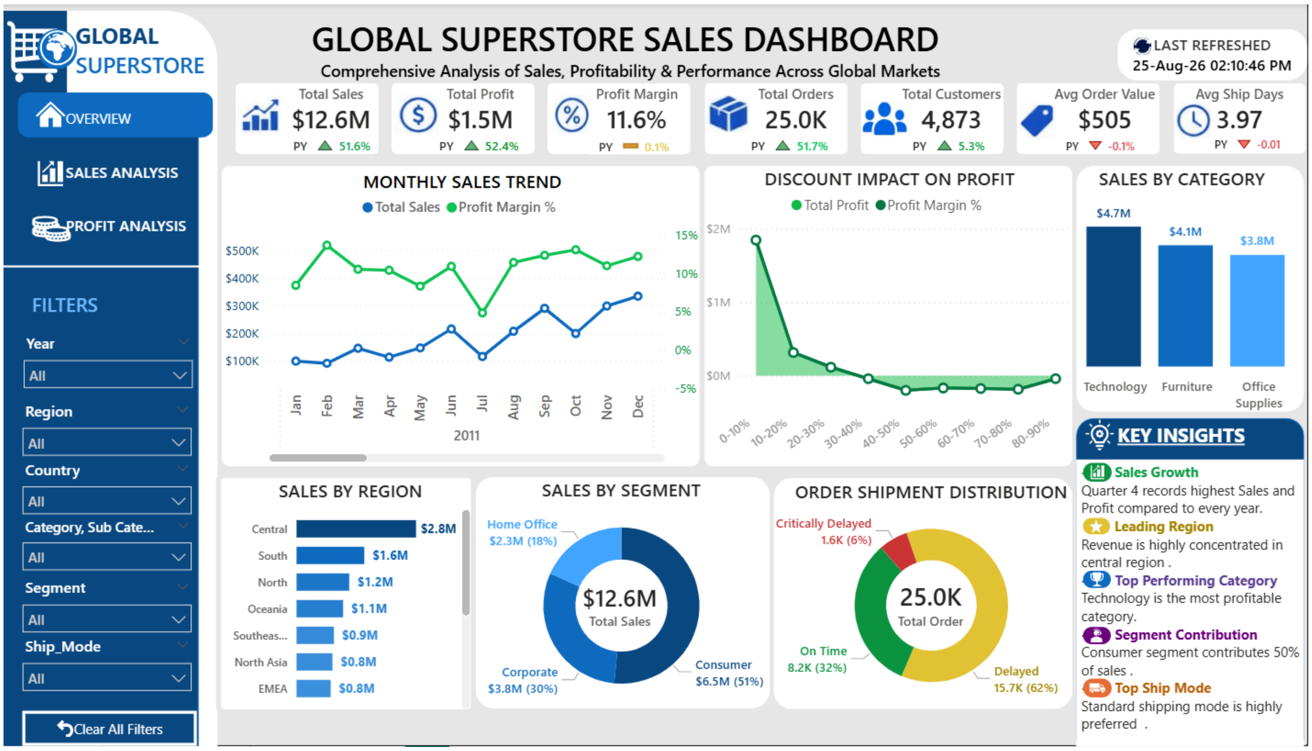
# 11. Power BI Dashboard Architecture

The supplied PBIX contains three report pages: OVERVIEW, SALES ANALYSIS and PROFIT ANALYSIS. The dashboard evidence confirms that the final interactive reporting layer is implemented in Power BI and uses common filters across the analytical pages.

| Page | Primary purpose | Major analytical components observed |
|---|---|---|
| OVERVIEW | Executive performance monitoring | KPI cards, monthly sales trend, discount impact, category sales, region sales, segment mix, shipment distribution, key insights |
| SALES ANALYSIS | Revenue and commercial drivers | monthly sales trend, country map, monthly/category matrix, sub-category ranking, ship-mode sales, top products, segment mix |
| PROFIT ANALYSIS | Profitability and margin drivers | Discount impact, profit by country, profit by ship mode, top sub-categories, monthly profit trend, profit by segment, month/category profit matrix |

The three-page structure creates a clear progression: establish the business position, investigate sales drivers, then investigate profitability drivers. This reduces the need to overload one canvas with every analytical question.

# 12. Dashboard Page 1 — Overview




*Supplied Power BI Overview page evidence.*

## Page Purpose

The Overview page acts as the executive entry point. It places the principal KPIs at the top and combines trend, category, region, segment, discount and shipment views underneath. The page also includes slicers for Year, Region, Country, Category/Sub-category, Segment and Ship Mode.

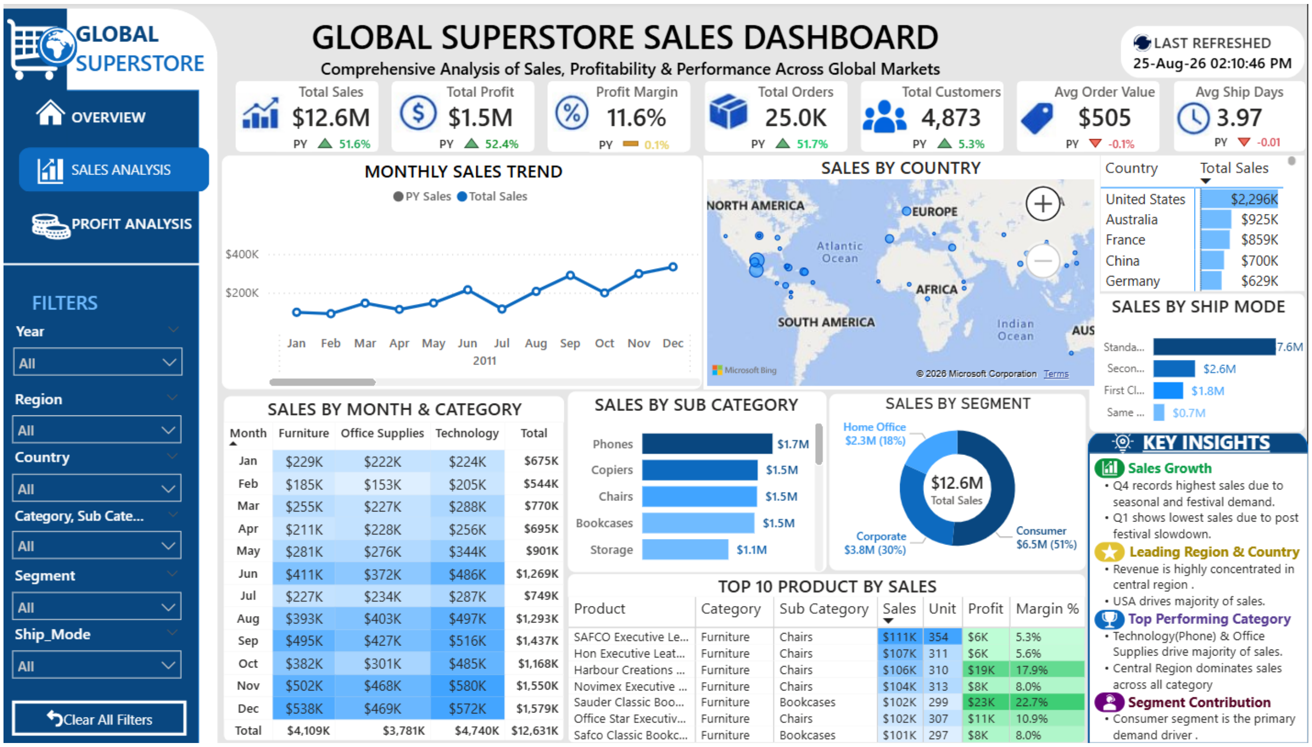
## Observed KPI Layer

| KPI card | Observed dashboard value |
|---|---|
| Total Sales | $12.6M |
| Total Profit | $1.5M |
| Profit Margin | 11.6% |
| Total Orders | 25.0K |
| Total Customers | 4,873 |
| Avg Order Value | $505 |
| Avg Ship Days | 3.97 |

## Interpretive Role

- KPI cards establish the overall commercial baseline.
- The monthly trend provides temporal context.
- Category and region visuals identify major revenue concentrations.
- Segment and shipment distributions provide customer and operational context.
- The discount-impact view connects commercial activity with profitability.

# 13. Dashboard Page 2 — Sales Analysis




*Supplied Power BI Sales Analysis page evidence.*

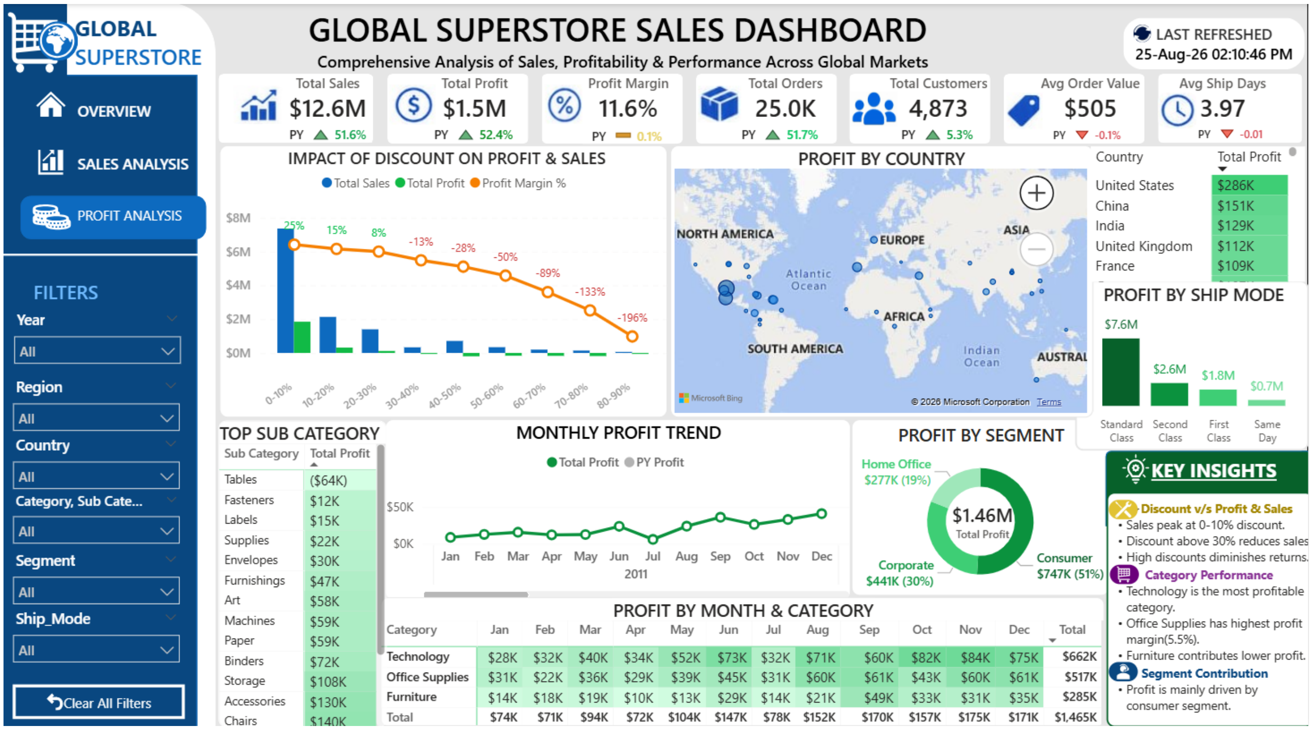
## Page Purpose

The Sales Analysis page moves from headline KPIs into revenue composition and sales drivers. It combines time, geography, category, sub-category, shipping mode, product and segment perspectives.

## Observed Visual Framework

| Visual | Decision question |
|---|---|
| Monthly Sales Trend | How does sales performance move through the year? |
| Sales by Country | Where is revenue geographically concentrated? |
| Sales by Month & Category | How do category contributions change across months? |
| Sales by Sub-Category | Which sub-categories drive revenue? |
| Sales by Ship Mode | Which shipping modes carry the largest sales volume? |
| Top 10 Product by Sales | Which products contribute the greatest sales? |
| Sales by Segment | Which customer segment contributes most revenue? |

# 14. Dashboard Page 3 — Profit Analysis




*Supplied Power BI Profit Analysis page evidence.*

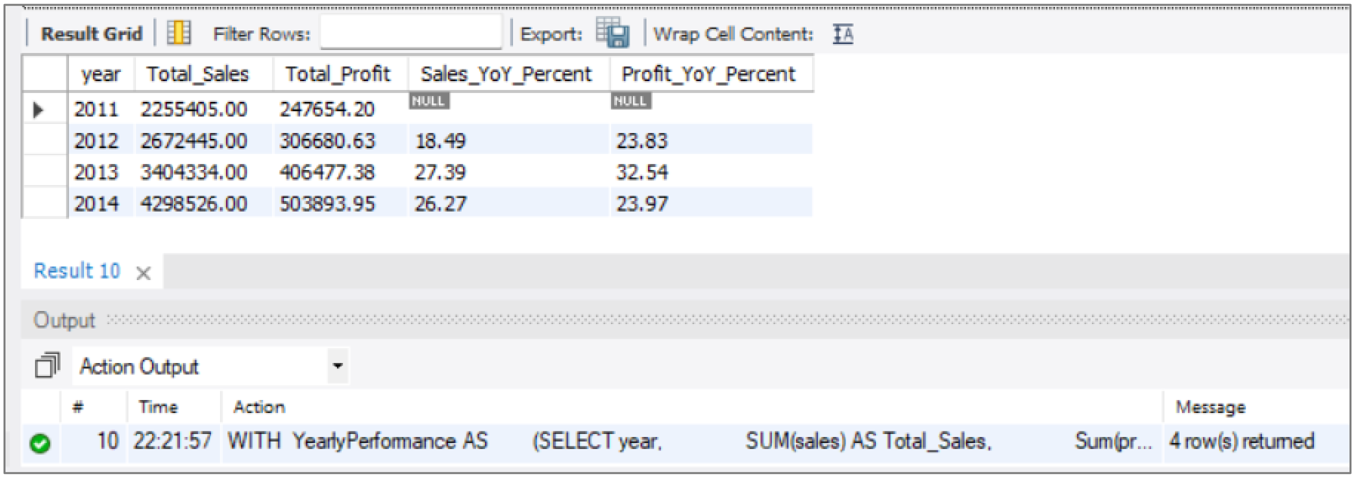
## Page Purpose

The Profit Analysis page isolates profitability from revenue scale. This is important because high sales do not necessarily imply strong economic performance.

## Observed Visual Framework

| Visual | Decision question |
|---|---|
| Impact of Discount on Profit & Sales | How does discount intensity affect economic performance? |
| Profit by Country | Where is profit concentrated geographically? |
| Profit by Ship Mode | How is profit distributed across delivery methods? |
| Top Sub-Category | Which sub-categories contribute strongest profit? |
| Monthly Profit Trend | How does profit evolve through the year? |
| Profit by Segment | Which segments contribute profit? |
| Profit by Month & Category | Which category-month combinations are most/least profitable? |

# 15. Business Performance Analysis




| year | Total_Sales | Total_Profit | Sales_YoY_Percent | Profit_YoY_Percent |
|---|---|---|---|---|
| 2011 | 2255405.00 | 247654.20 | NULL | NULL |
| 2012 | 2672445.00 | 306680.63 | 18.49 | 23.83 |
| 2013 | 3404334.00 | 406477.38 | 27.39 | 32.54 |
| 2014 | 4298526.00 | 503893.95 | 26.27 | 23.97 |

*Yearly Sales Performance from Independent SQL Validation.*

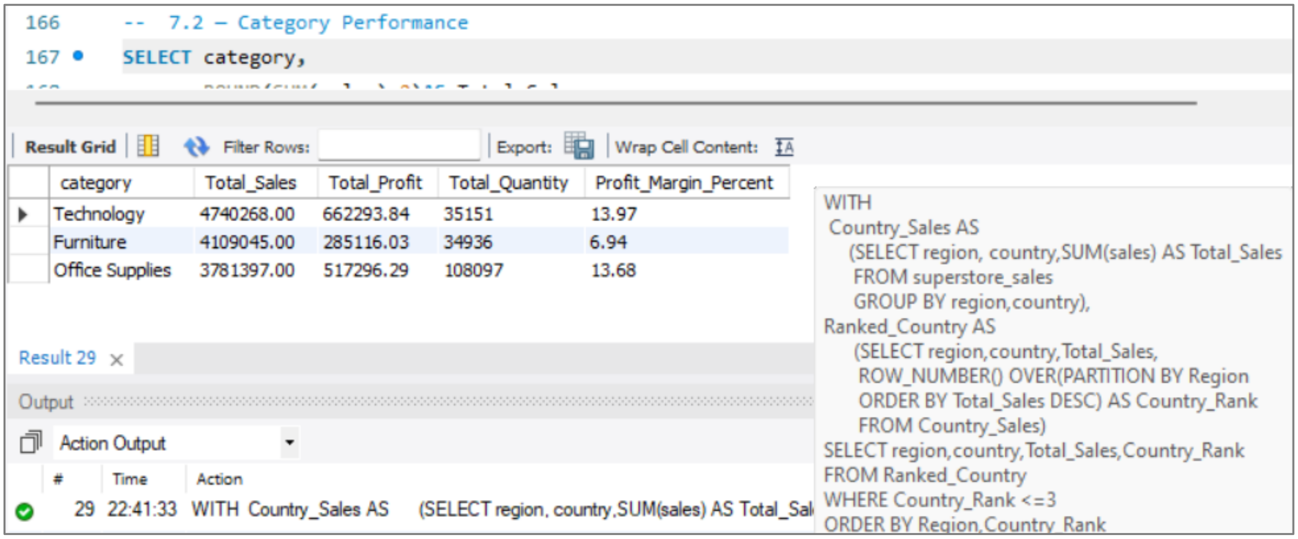
| Year | Sales | Profit | Sales YoY | Profit YoY |
|---|---|---|---|---|
| 2011 | $2.26M | $248K | — | — |
| 2012 | $2.67M | $307K | 18.49% | 23.83% |
| 2013 | $3.40M | $406K | 27.39% | 32.54% |
| 2014 | $4.30M | $504K | 26.27% | 23.97% |

Sales increased every year from $2.26M in 2011 to $4.30M in 2014. The strongest annual sales growth in this four-year sequence occurred in 2013 at 27.39%, followed by 26.27% in 2014.

Profit followed the same upward direction, increasing from $248K to $504K. Profit growth was strongest in 2013 at 32.54%.

Quarterly analysis also shows Q4 as the largest quarter by both sales and profit in the supplied data, with approximately $4.30M in sales and $0.50M in profit.

# 16. Product and Category Analysis




*Sales contribution by major product category from Independent SQL Validation.*

| Category | Sales | Profit | Margin | Quantity |
|---|---|---|---|---|
| Technology | $4.74M | $662K | 13.97% | 35,151 |
| Furniture | $4.11M | $285K | 6.94% | 34,936 |
| Office Supplies | $3.78M | $517K | 13.68% | 108,097 |

Technology is the largest category by sales at $4.74M and also has the highest category profit at $662K. Its margin is approximately 13.97%.

Office Supplies produces $517K of profit despite lower sales than Technology. Furniture generates $4.11M of sales but has the lowest category margin at 6.94%.

## Sub-Category Highlights

| Sub-category | Sales | Profit | Margin |
|---|---|---|---|
| Phones | $1.70M | $215K | 12.65% |
| Copiers | $1.51M | $259K | 17.13% |
| Chairs | $1.50M | $140K | 9.36% |
| Bookcases | $1.47M | $162K | 11.04% |
| Storage | $1.13M | $108K | 9.61% |
| Appliances | $1.01M | $141K | 14.02% |
| Machines | $779K | $59K | 7.54% |
| Tables | $757K | $-64K | -8.47% |
| Accessories | $749K | $130K | 17.30% |
| Binders | $462K | $72K | 15.67% |

Phones lead sub-categories by sales at $1.70M, while Copiers produce the highest profit among the listed sub-categories at $259K. Tables are a major profitability concern, with a negative total profit of $64K.

# 17. Customer and Segment Analysis

| Segment | Sales | Profit | Orders | Margin |
|---|---|---|---|---|
| Consumer | $6.50M | $747K | 13,104 | 11.50% |
| Corporate | $3.82M | $441K | 7,673 | 11.53% |
| Home Office | $2.31M | $277K | 4,687 | 11.99% |

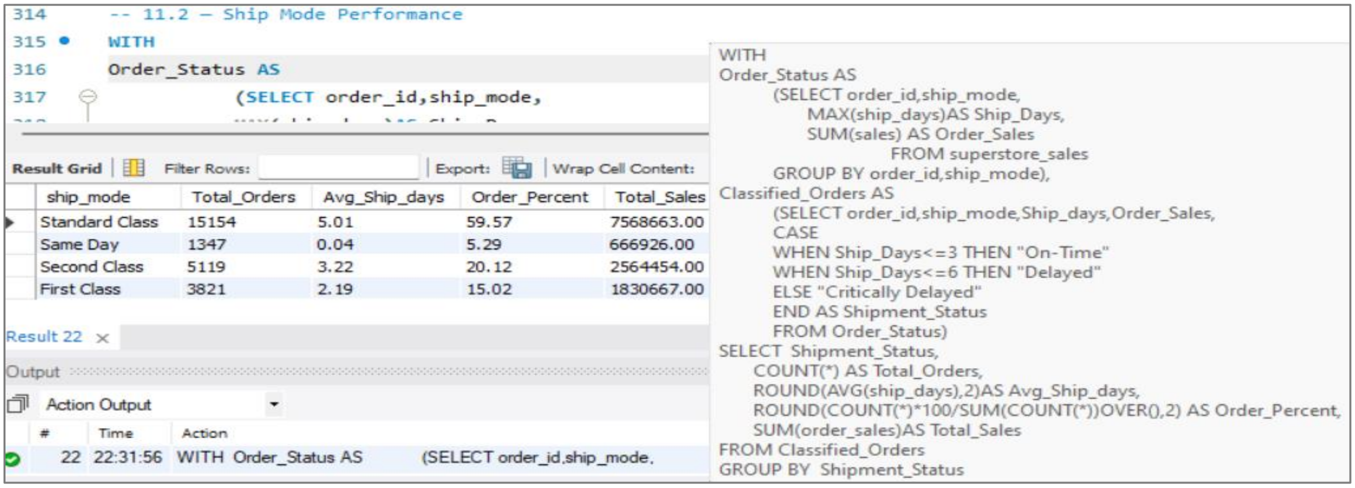
Consumer is the largest segment by sales at $6.50M and contributes $747K in profit. Corporate follows with $3.82M in sales. Home Office has the highest segment margin at approximately 11.99%.

## Customer Concentration

| Customer | Sales | Profit | Orders |
|---|---|---|---|
| Sean Miller (SM-203204) | $25K | $-2K | 5 |
| Tamara Chand (TC-209804) | $19K | $9K | 5 |
| Cari Sayre (CS-118451) | $17K | $3K | 11 |
| Susan Pistek (SP-209202) | $17K | $5K | 12 |
| Vivek Grady (VG-218051) | $16K | $3K | 7 |
| Raymond Buch (RB-193604) | $15K | $7K | 6 |
| Peter Fuller (PF-191201) | $15K | $1K | 8 |
| Tom Ashbrook (TA-213854) | $15K | $5K | 4 |
| Barry Franz (BF-110051) | $15K | $2K | 9 |
| Adrian Barton (AB-101054) | $14K | $5K | 10 |

The customer-level structure supports concentration analysis through SQL and Power BI. This is useful for identifying high-value accounts, prioritizing retention activity and monitoring whether revenue concentration is accompanied by healthy profitability.

# 18. Shipping and Operational Analysis




*Shipping and Operational Analysis from Independent SQL Validation.*

| Ship Mode | Sales | Profit | Orders | Avg Ship Days | Margin |
|---|---|---|---|---|---|
| Standard Class | $7.57M | $888K | 15,154 | 5.00 | 11.74% |
| Second Class | $2.56M | $292K | 5,119 | 3.23 | 11.39% |
| First Class | $1.83M | $208K | 3,821 | 2.18 | 11.36% |
| Same Day | $667K | $76K | 1,347 | 0.04 | 11.40% |

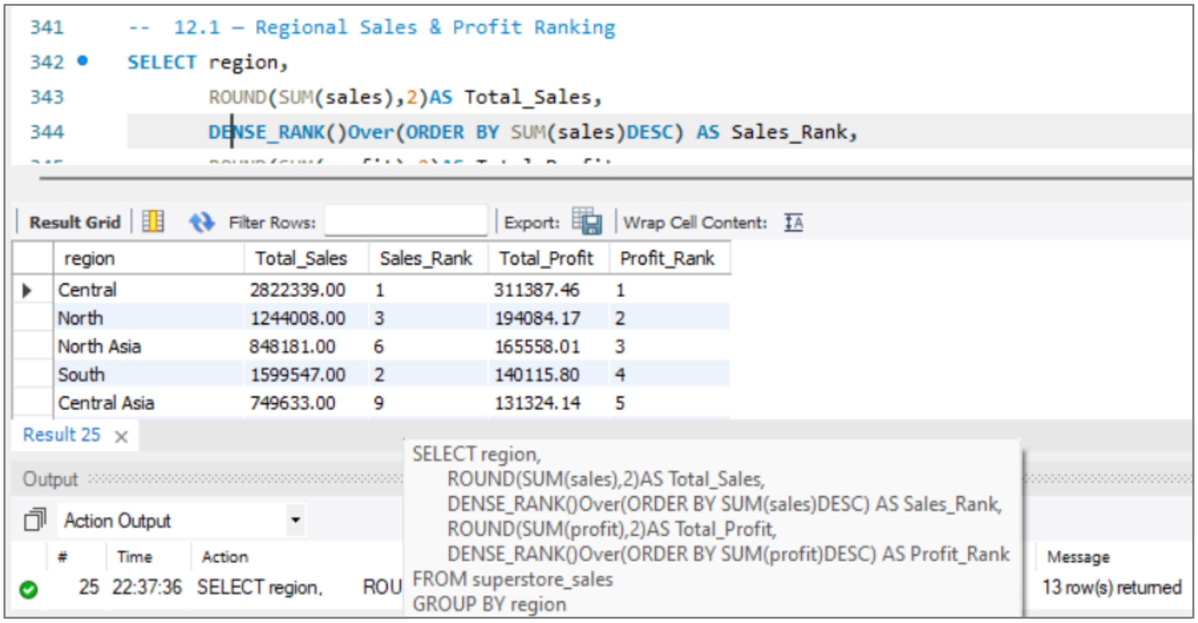
## Order-Level Shipment Status

| Shipment status | Orders | Order % | Sales | Avg Days |
|---|---|---|---|---|
| On-Time | 8,210 | 32.3% | $4.05M | 1.90 |
| Delayed | 15,677 | 61.6% | $7.78M | 4.75 |
| Critically Delayed | 1,554 | 6.1% | $804K | 7.00 |

Standard Class carries the largest commercial volume, with approximately $7.57M in sales and 15,154 distinct orders. The order-level classification shows that delayed shipments represent the largest shipment-status group under the project's rule set.

The SQL validation script classifies order-level shipment performance using maximum Ship_Days per Order_ID and assigns On-Time for up to 3 days, Delayed for 4–6 days and Critically Delayed beyond 6 days. This is an important methodological distinction from simply classifying individual rows.

# 19. Geographic Analysis




*Top regions by sales from the Independent SQL Validation.*

| Region | Sales | Profit | Margin |
|---|---|---|---|
| Central | $2.82M | $311K | 11.03% |
| South | $1.60M | $140K | 8.76% |
| North | $1.24M | $194K | 15.60% |
| Oceania | $1.10M | $120K | 10.91% |
| Southeast Asia | $883K | $18K | 1.99% |
| North Asia | $848K | $166K | 19.52% |
| EMEA | $806K | $44K | 5.44% |
| Africa | $783K | $89K | 11.32% |
| Central Asia | $750K | $131K | 17.52% |
| West | $725K | $108K | 14.94% |
| East | $678K | $91K | 13.47% |
| Caribbean | $324K | $35K | 10.66% |
| Canada | $67K | $18K | 26.62% |

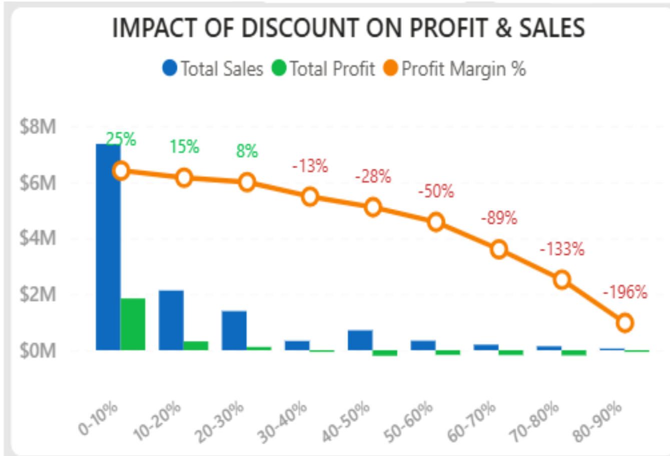
Central is the leading region by sales at $2.82M, followed by South at $1.60M. North has a comparatively strong margin of 15.60%, while Southeast Asia has a much lower margin of 1.99%.

## Top Countries by Sales

| Country | Sales | Profit | Margin |
|---|---|---|---|
| United States | $2.30M | $286K | 12.46% |
| Australia | $925K | $104K | 11.22% |
| France | $859K | $109K | 12.69% |
| China | $700K | $151K | 21.51% |
| Germany | $629K | $107K | 17.07% |
| Mexico | $619K | $102K | 16.53% |
| India | $590K | $129K | 21.89% |
| United Kingdom | $528K | $112K | 21.16% |
| Indonesia | $405K | $16K | 3.86% |
| Brazil | $361K | $30K | 8.33% |

The geographic layer is designed for both revenue concentration and profitability screening. The SQL script further supports ranking the top three countries within each region, which provides a more actionable regional drill-down than a single global country ranking.

# 20. Discount and Profitability Analysis




*Profit margin by discount range, calculated from the power BI visual.*

| Discount range | Sales | Profit | Margin | Quantity |
|---|---|---|---|---|
| 0-10% | $7.37M | $1.85M | 25.08% | 100,870 |
| 10-20% | $2.13M | $314K | 14.74% | 20,143 |
| 20-30% | $1.40M | $116K | 8.31% | 21,049 |
| 30-40% | $330K | $-42K | -12.72% | 2,067 |
| 40-50% | $710K | $-200K | -28.16% | 15,978 |
| 50-60% | $338K | $-168K | -49.63% | 6,291 |
| 60-70% | $196K | $-175K | -89.26% | 5,605 |
| 70-80% | $140K | $-186K | -133.33% | 4,928 |
| 80-90% | $21K | $-42K | -196.08% | 1,253 |

The relationship between discount intensity and profitability is pronounced in the supplied data. The 0–10% range has a margin of approximately 25.08%, while progressively higher discount ranges move into negative profitability.

Discount ranges from 30–40% onward are loss-making in aggregate. The 80–90% range has the most negative margin at approximately -196.08%. This pattern makes discount governance one of the strongest commercial-control opportunities in the dataset.

## Profit Status Distribution

| Profit status | Records | Sales | Profit |
|---|---|---|---|
| Break Even | 668 | $153K | $0K |
| Exceptional Record | 1 | $0K | $-0K |
| Loss | 12,540 | $2.49M | $-921K |
| Profit | 38,046 | $9.99M | $2.39M |

The prepared dataset contains 38,046 profit records, 12,540 loss records and 668 break-even records. One exceptional record has zero sales and a negative profit value; this is consistent with the missing profit-margin value discussed in the data-quality section.

# 21. Year-over-Year Performance

Year-over-year validation is implemented in SQL using a yearly aggregation followed by LAG() to retrieve the previous year's sales and profit. The percentage change is then calculated against the prior-year value with NULLIF protection against division by zero.

| Year | Sales | Sales YoY | Profit | Profit YoY |
|---|---|---|---|---|
| 2011 | $2.26M | — | $248K | — |
| 2012 | $2.67M | 18.49% | $307K | 23.83% |
| 2013 | $3.40M | 27.39% | $406K | 32.54% |
| 2014 | $4.30M | 26.27% | $504K | 23.97% |

## Interpretation

- 2012 shows positive growth in both sales and profit over 2011.
- 2013 records the strongest sales growth of the four-year sequence and the strongest profit growth.
- 2014 continues the positive growth trend, with sales growth of approximately 26.27% and profit growth of approximately 23.97%.
- The direction of growth is consistent across the annual series, supporting the use of YoY measures as a dashboard monitoring mechanism.

# 22. SQL Independent Validation Framework

SQL is used as an independent validation layer. The supplied script is organized into setup, data import, data-quality validation, performance optimization, KPI analysis, time analysis, product/category analysis, customer analysis, segment analysis, discount analysis, shipping analysis, regional analysis and YoY validation.

## Validation Philosophy

- Recalculate selected metrics outside the Power BI visual layer.
- Use the same analytical definitions where direct reconciliation is intended.
- Validate structural conditions before interpreting business results.
- Use SQL window functions for ranking and prior-year comparisons.
- Keep the validation process reproducible through named query sections.

## Observed SQL Query Groups

| Group | SQL area | Purpose |
|---|---|---|
| 01 | Database & Table Setup | Create database and define superstore_sales schema |
| 02 | Data Import & MySQL Configuration | Configure local import and load cleaned CSV |
| 03 | Data Structure & Quality Validation | Zero sales, date integrity, shipping days and duplicate controls |
| 04 | Query Performance Optimization | Indexes for country, region, category and order date |
| 05 | KPI & Overall Business Performance | Sales, profit, margin, orders, customers, AOV and shipping time |
| 06 | Time & Trend Analysis | Monthly sales, profit and margin |
| 07 | Product & Category Performance | Profit status, category, sub-category, product and loss-product ranking |
| 08 | Customer Performance | Customer metrics, above-average customers and cumulative sales concentration |
| 09 | Customer Segment Performance | Segment sales, profit, quantity and margin |
| 10 | Discount Analysis | Discount-range sales, profit, quantity and margin |
| 11 | Shipping & Operational Performance | Shipment status and ship-mode performance |
| 12 | Regional & Geographical Performance | Region rankings and top countries within regions |
| 13 | Year-over-Year Business Performance Validation | Annual sales/profit and LAG-based YoY calculations |

# 23. Validation Results and Reconciliation

The independent SQL layer provides direct numerical definitions that can be reconciled across the cleaned workbook and the Power BI KPI layer. The most important validation target is not visual appearance; it is agreement of underlying business definitions and totals.

## Core Metric Reconciliation

| Metric | Cleaned data calculation | Power BI dashboard evidence | SQL definition |
|---|---|---|---|
| Total Sales | $12.63M | $12.6M | SUM(Sales) |
| Total Profit | $1.46M | $1.5M | SUM(Profit) |
| Profit Margin | 11.60% | 11.6% | SUM(Profit) / SUM(Sales) |
| Orders | 25,035 | 25.0K | COUNT(DISTINCT Order_ID) |
| Customers | 4,873 | 4,873 | COUNT(DISTINCT Customer_ID) |
| Avg Order Value | $1K | $505 | SUM(Sales) / COUNT(DISTINCT Order_ID) |
| Avg Ship Days | 3.97 | 3.97 | AVG(Ship_Days) |

The rounded dashboard values are consistent with the independently calculated cleaned-data totals. Small presentation differences are expected because the dashboard uses abbreviated display formats such as $12.6M and 25.0K.

## Validation Scope

- Core KPI totals are directly reproducible.
- Monthly and annual trends can be reproduced through GROUP BY time fields.
- Category, sub-category, product, customer, segment, ship mode and region summaries are reproducible from the single analytical table.
- YoY values are independently generated through LAG() rather than copied from Power BI.
- Data-quality checks are represented in SQL as executable controls.

# 24. Key Business Insights

**1. Sustained growth:** Sales increased from $2.26M in 2011 to $4.30M in 2014, with positive annual growth in every comparable year.

**2. Technology is a major commercial engine:** Technology contributes $4.74M in sales and $662K in profit, with a margin of 13.97%.

**3. Furniture needs profitability attention:** Furniture has $4.11M in sales but only $285K in profit, producing the lowest category margin at 6.94%.

**4. Consumer is the largest segment:** Consumer contributes $6.50M in sales and $747K in profit.

**5. Discount intensity is a strong risk signal:** Margins fall from 25.08% in the 0–10% range to negative territory from the 30–40% range onward.

**6. Standard Class dominates shipping volume:** Standard Class carries $7.57M in sales and 15,154 orders.

**7. Central leads regional sales:** Central is the largest region at $2.82M in sales, while North and North Asia show stronger margins than several high-volume regions.

**8. Tables are a notable loss area:** The Tables sub-category records approximately $64K of negative profit.

**9. Q4 is the strongest quarter:** Q4 contributes approximately $4.30M in sales, the highest quarterly total.

**10. Data quality is largely controlled:** The cleaned table has no remaining Order–Product–Customer duplicates, no invalid order/ship date sequences and no negative Ship_Days.

# 25. Recommendations

| Recommendation | Action | Priority |
| --- | --- | --- |
| Discount governance | Set stronger approval thresholds for high-discount transactions, particularly above 30%, where aggregate profitability becomes negative. | High |
| Furniture profitability | Review pricing, discounting, shipping cost and product-level economics within Furniture, with special attention to loss-making sub-categories such as Tables. | High |
| Product portfolio review | Investigate recurring loss-making products and distinguish structural low-margin products from one-off exceptions. | High |
| Regional profitability | Use regional margin alongside sales volume so commercial teams do not optimize revenue at the expense of contribution. | High |
| Shipping policy | Monitor delayed and critically delayed order groups and evaluate whether high-volume shipping modes can be optimized without harming service levels. | Medium |
| Customer prioritization | Combine customer sales, profit and order frequency to identify high-value accounts and accounts with high revenue but weak contribution. | Medium |
| Quarter-end planning | Prepare inventory, fulfillment and commercial campaigns ahead of Q4 because it is the strongest sales quarter in the supplied data. | Medium |
| Validation controls | Retain SQL KPI and YoY validation as a repeatable QA step whenever the Power BI model or Power Query transformations are changed. | High |

# 26. Limitations and Analytical Controls

## Known limitations

- The analysis is based on the fields and business definitions present in the supplied dataset.
- Profitability analysis is descriptive; it does not establish causal effects of discounts without controlled experimentation.
- The dashboard's abbreviated values are presentation formats; detailed reconciliation should use underlying values.
- One zero-sales record creates an undefined conventional profit-margin calculation and should remain treated as a data-quality exception.
- The PBIX package exposes model/report metadata and measure names, but not every internal DAX expression is recoverable from the supplied package in plain text. The report therefore avoids inventing formulas not directly evidenced by the source files.

## Controls that strengthen reliability

- Power Query provides a repeatable transformation chain.
- Composite duplicate removal is explicitly documented.
- Date and shipping-duration quality checks are executable in SQL.
- Core KPI definitions are reproduced independently in SQL.
- Annual YoY calculations are independently generated using LAG().
- Dashboard evidence is retained as page-level visual documentation.

# 27. Conclusion

The Global Superstore Sales Analysis project establishes a complete analytical workflow from Excel source extraction through Power Query preparation, Power BI modeling and dashboarding, and independent SQL validation. The resulting analytical dataset contains 51,255 cleaned records and supports a broad commercial view across time, products, customers, shipping and geography.

The business picture is characterized by sustained sales growth, strong contribution from Technology and Consumer, a meaningful profitability gap within Furniture, and a clear deterioration in margin as discount intensity increases. Operationally, Standard Class dominates volume, while regional profitability varies considerably.

The strongest feature of the project is the separation of responsibilities between tools: Power Query handles preparation, Power BI handles interactive analytics, and SQL provides an independent control layer. This makes the project suitable as a portfolio example of practical data preparation, BI modeling, dashboard development and analytical validation.

# 28. Technical Skills Demonstrated

| Skill area | Evidence in project |
|---|---|
| Excel data handling | Source workbook organization, structured analytical tables and supporting summaries |
| Power Query | Repeatable cleaning, data typing, text cleanup, duplicate removal and derived features |
| Data quality | Duplicate, date, shipping, zero-sales and missing-value controls |
| Data modeling | Fact and dimension structures in Power BI |
| DAX / measures | Core KPIs, prior-year metrics and YoY measures |
| Dashboard design | Three-page interactive Power BI report with slicers, KPI cards, charts, matrices and maps |
| SQL | Schema setup, import, quality validation, aggregations, CTEs, window functions and ranking |
| Business analysis | Sales, profit, margin, product, customer, segment, discount, shipping and geographic interpretation |
| Validation | Independent reproduction of core metrics and YoY calculations |
| Communication | Executive insights and action-oriented recommendations |

# Appendix A — Data Dictionary

| Field | Definition | Type / role |
|---|---|---|
| Order_ID | Transaction/order identifier | Identifier |
| Order_Date | Date the order was placed | Date |
| Ship_Date | Date the order was shipped | Date |
| Ship_Mode | Shipping method | Category |
| Order_Priority | Order priority level | Category |
| Customer_ID | Customer identifier | Identifier |
| Customer_Name | Customer name | Text |
| Segment | Customer segment | Category |
| Product_ID | Product identifier | Identifier |
| Product_Name | Product description/name | Text |
| Category | High-level product category | Category |
| Sub_Category | Product sub-category | Category |
| Market | Market grouping | Category |
| Region | Geographic region | Category |
| Country | Country | Geography |
| State | State/province | Geography |
| City | City | Geography |
| Sales | Sales amount | Numeric |
| Quantity | Units sold | Numeric |
| Discount | Discount rate/value | Numeric |
| Discount_Range | Derived discount band | Derived |
| Discount_Category | Derived discount classification | Derived |
| Profit | Profit amount | Numeric |
| Profit_Margin_Percent | Profit as a percentage of sales | Derived |
| Profit_Status | Profitability classification | Derived |
| Ship_Cost | Shipping cost | Numeric |
| Ship_Days | Derived shipping duration | Derived |
| Ship_Status | Derived shipping classification | Derived |
| Year | Order year | Derived |
| Quarter | Order quarter | Derived |
| Month_No | Month number | Derived |
| Month Name | Month name | Derived |

# Appendix B — SQL Query Inventory

The supplied SQL script contains a structured validation and analysis sequence. The inventory below documents the purpose of each query group so that the script can be maintained as a professional validation artifact.

| Group | Queries / scope | Analytical purpose |
|---|---|---|
| 01 | CREATE DATABASE, USE, CREATE TABLE | Establish analytical environment and schema |
| 02 | MySQL configuration, LOAD DATA, structure preview, row count | Load and inspect the prepared dataset |
| 03 | Zero sales, date checks, shipping checks, duplicate checks | Structural and data-quality validation |
| 03 continued | Row ID and uniqueness constraint | Enforce transaction-level uniqueness |
| 04 | Indexes on Country, Region, Category, Order_Date | Support query performance |
| 05 | Overall KPI aggregation | Validate sales, profit, margin, orders, customers, AOV and shipping |
| 06 | Monthly trend | Validate time-based sales, profit and margin |
| 07 | Profit status, category, sub-category, product, loss-product ranking | Product and category performance |
| 08 | Customer metrics, above-average customers, cumulative contribution | Customer performance and concentration |
| 09 | Segment aggregation | Segment contribution and profitability |
| 10 | Discount range aggregation | Discount economics |
| 11 | Order-level shipment classification and ship-mode analysis | Operational performance |
| 12 | Region ranking and top countries within region | Geographic performance |
| 13 | Annual aggregation with LAG() | Independent YoY validation |

## Selected Validation SQL Patterns

```
-- Overall KPI validation
SELECT
    ROUND(SUM(sales),2) AS total_sales,
    ROUND(SUM(profit),2) AS total_profit,
    ROUND(SUM(profit)/SUM(sales),2) AS profit_margin,
    COUNT(DISTINCT order_id) AS total_orders,
    COUNT(DISTINCT customer_id) AS total_customers
FROM superstore_sales;

-- YoY validation pattern
LAG(Total_Sales)  OVER (ORDER BY year) AS Previous_Year_Sales,
LAG(Total_Profit) OVER (ORDER BY year) AS Previous_Year_Profit
```

# Appendix C — Evidence Register

The report was prepared from the project artifacts supplied in the working session. The register below identifies how each artifact contributes to the documented analysis.

| Artifact | Used for |
|---|---|
| Global_Superstore_Cleaned_Data.xlsx | Workbook structure, cleaned dataset, field definitions and quantitative calculations |
| Global_Superstore_PBIX.pbix | Power BI page names, model objects, measure inventory and report layout |
| Global_Superstore_SQL_Validation.sql | SQL validation methodology, query grouping and independent metric definitions |
| Power Query evidence | 04_Power_Query_Data_Cleaning.png — Power Query Applied Steps and cleaned-data evidence |
| Dashboard screenshots | 01_Dashboard_Overview.png<br>02_Dashboard_Sales_Analysis.png<br>03_Dashboard_Profit_Analysis.png |

## Final Analytical Deliverable

The report is intended to accompany the Power BI PBIX, cleaned Excel workbook and SQL validation script as a single portfolio-ready project package. Each layer has a defined purpose and the analytical conclusions are grounded in the supplied project data and artifacts.